Rotary Wing Aerodynamics (AE 667)



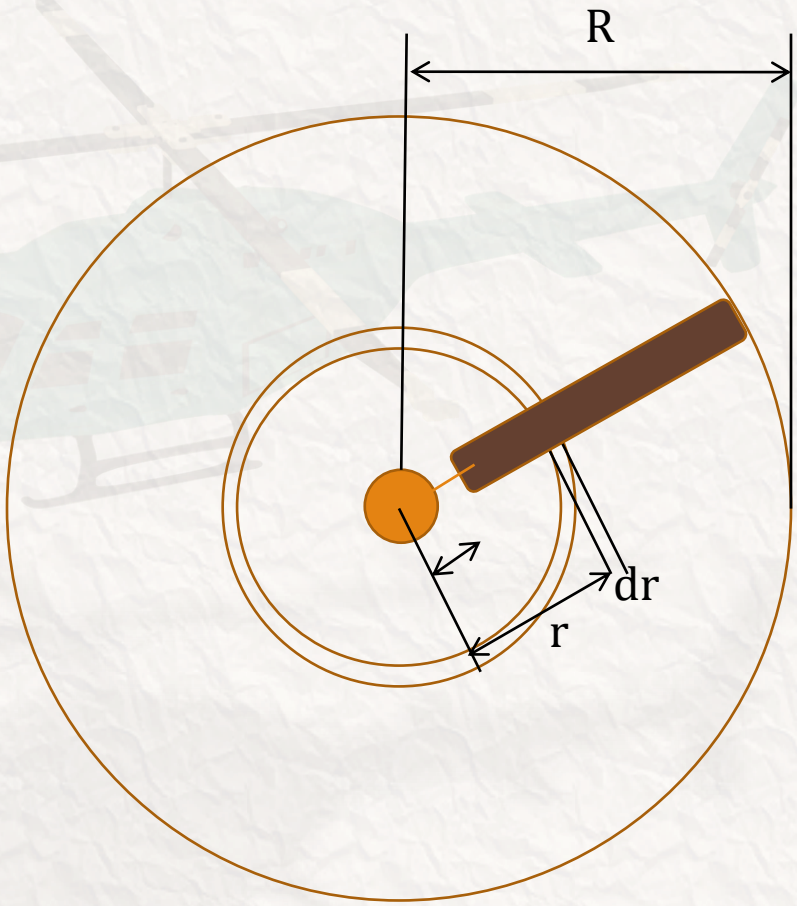
Blade Element Momentum (BEM) Theory

Background

- Addresses uniform inflow assumption
  - Uniform inflow results in lowest power consumption, hence BE theory under-predicts power
- Combination of Momentum and Blade Element Theories

Procedure…

- Apply Momentum Theory on a strip of the disk
- Apply BE theory on the strips of the blades

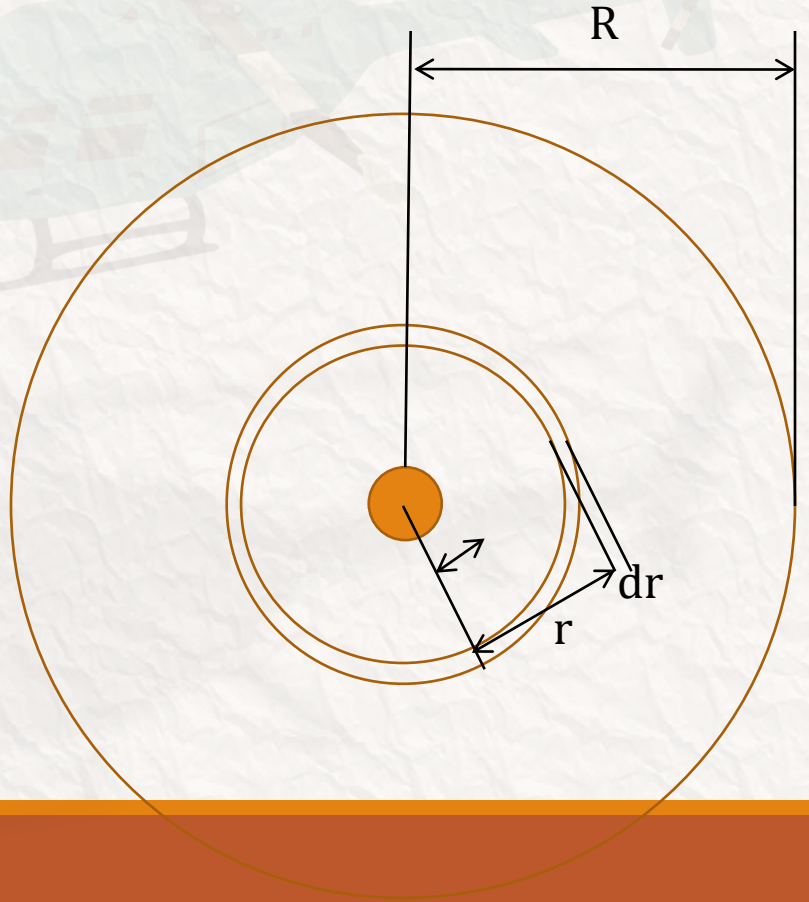


Momentum Theory side

- Consider annular ring element of *dr* thickness
- Area = 2πrdr
- Mass flow rate ($\dot{dm}$) = 2πρr(V+v)dr
- Thrust (dT) $= \dot{dm}v_4 = 2\dot{dm}v$

  [From momentum theory]

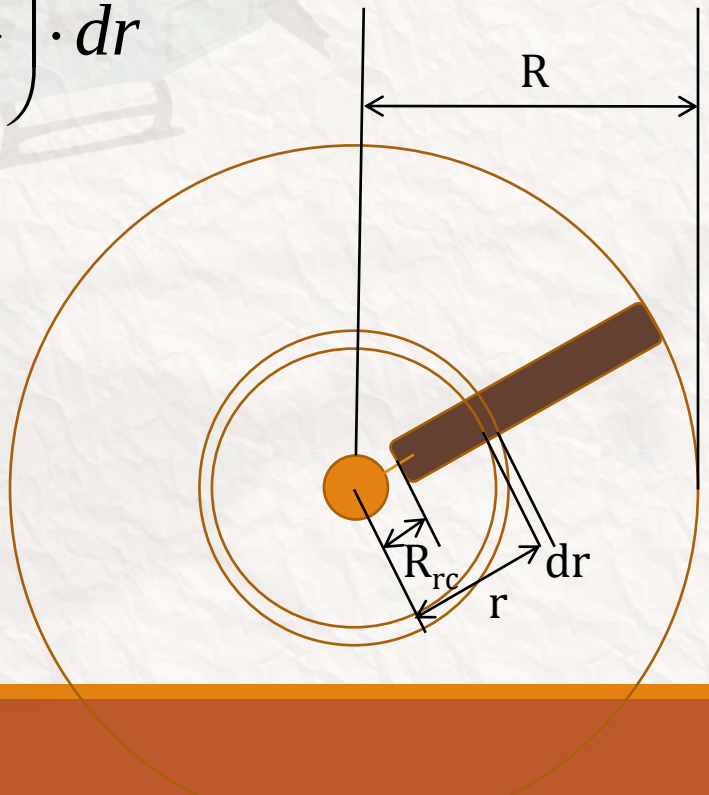
  dT = 4πρr(V+v)vdr



Blade Element Theory side

$$dT = b \cdot \frac{1}{2} \rho \cdot (\Omega r)^2 c \cdot C_l \cdot dr$$

$$= abc \cdot \frac{1}{2} \rho \cdot (\Omega r)^2 \cdot \left(\theta - \frac{V + v}{\Omega r} \right) \cdot dr$$



Combining the two

- Equating the thrusts (dT) from both approaches, we get:

$$\lambda^2 + \left(\frac{\sigma a}{8} - \lambda_c\right)\lambda - \frac{\sigma a}{8}\theta\frac{r}{R} = 0$$

or

$$\lambda = \sqrt{\left(\frac{\sigma a}{16} - \frac{\lambda_c}{2}\right)^2 + \frac{\sigma a}{8}\theta\frac{r}{R}} - \left(\frac{\sigma a}{16} - \frac{\lambda_c}{2}\right)$$

where,

$$\lambda_c = \frac{V}{\Omega R}; \quad \lambda = \frac{V + \mathrm{v}}{\Omega \mathrm{R}}$$

- [Be mindful about the definition of σ in this context!]

Integrating (same as BE Theory)

$$T = b\int_{R_{RC}}^{R} dT\,;$$

$$= b\int_{R_{RC}}^{R} \frac{1}{2}\rho\left(U_T^2 + U_P^2\right)c\left(C_l\cos(\phi) - C_d\sin(\phi)\right)dr$$

$$Q = b\int_{R_{RC}}^{R} rdF_x \quad \text{(Torque)}$$

$$= b\int_{R_{RC}}^{R} r\frac{1}{2}\rho\left(U_T^2 + U_P^2\right)c\left(C_d\cos(\phi) + C_l\sin(\phi)\right)dr$$

$$P = b\int_{R_{RC}}^{R} \Omega r dF_X$$

$$= b\int_{R_{RC}}^{R} \Omega r\frac{1}{2}\rho\left(U_T^2 + U_P^2\right)c\left(C_d\cos(\phi) + C_l\sin(\phi)\right)dr$$

$$= Q\Omega$$

The only difference is, that now expressions for U_P and ϕ are changed

$$\lambda = \sqrt{\left(\frac{\sigma a}{16} - \frac{\lambda_c}{2}\right)^2 + \frac{\sigma a}{8}\theta\frac{r}{R}} - \left(\frac{\sigma a}{16} - \frac{\lambda_c}{2}\right)$$

Ideal Rotor

- Lowest induced drag through uniform inflow

- Uniform inflow possible theoretically through non-linear twist blades: $\theta = C(R/r)$.
  [when σ is constant (no taper)].

$$\lambda = \sqrt{\left(\frac{\sigma a}{16} - \frac{\lambda_c}{2}\right)^2 + \frac{\sigma a}{8}\theta\frac{r}{R}} - \left(\frac{\sigma a}{16} - \frac{\lambda_c}{2}\right)$$

Ideal Rotor

$$\lambda = \sqrt{\left(\frac{\sigma a}{16} - \frac{\lambda_c}{2}\right)^2 + \frac{\sigma a}{8}\theta\frac{r}{R}} - \left(\frac{\sigma a}{16} - \frac{\lambda_c}{2}\right)$$

- Very high pitch angles near root ➔ stall
- Also, such non-linear twist rotor is difficult to manufacture
- Ideas to address these problems?

Ideal Rotor

So Let's say we made an "Ideal Rotor"

Anything else one can wish for?

Optimum Rotor (Twist)

- Focus on minimizing total power for a given thrust.
- $FM = \frac{\lambda C_T}{k\lambda C_T + \sigma C_d/8} = \frac{1}{k + \frac{\sigma C_d}{8\lambda C_T}} = \frac{1}{k + \frac{3\,C_d}{4\lambda\,\bar{C}_l}}$ $using:\ \bar{C}_l = 6\frac{C_T}{\sigma}$
- The above expression implies that for an optimum performance, each blade section must operate at maximum $\frac{C_l}{C_d}$ (or $\frac{C_l^{3/2}}{C_d}$ for hover, when λ is taken as $\sqrt{\frac{C_T}{2}}$)
- Let the angle of attack corresponding to this be $\alpha_{Optimum}$
- Hence the pitch at a radial section should be:

$$\theta = \alpha_{Optimum} + \tan^{-1}\left(\frac{V + v}{\Omega r}\right)$$

Optimum Rotor (Taper)

- Expression for dT from Momentum and BE side as seen earlier:

$$dT = b\frac{1}{2}\rho(\Omega r)^2 c C_l dr = 4\pi\rho r v^2 dr$$

- For the case of optimally twisted rotor (C_l constant and optimum) with uniform inflow (v), the local solidity $\sigma(r)$ is given by:

$$\sigma(r) = \frac{bc}{\pi R} = \left(\frac{8v^2}{\Omega^2 R C_l}\right)\frac{1}{r} = \frac{Const}{r}$$

This gives optimal taper profile $c(r)$

Optimum Rotor

- Such rotors are difficult to make
- Best performance only at a given thrust setting
- Practically, **linear twist + linear taper** used for optimum performance

Accounting for losses not covered directly in BEM Theory

- Compressibility wave effects:
  - Values of C_l and C_d in the lookup table can be adjusted to account for compressibility effects
  - Prandtl – Glauert compressibility correction
  - $C_l = \frac{C_{l,incompressible}}{\sqrt{1-M_\infty^2}}$; $C_d = \frac{C_{d,incompressible}}{\sqrt{1-M_\infty^2}}$

- Tip losses (due to finiteness of wing/blade)
  - Prandtl's empirical tip loss function

Prandtl's Tip Loss Model

- Suggested multiplying sectional inflow (V+v) by a factor **F** in the thrust expression.

$$F = \frac{2}{\pi} Cos^{-1}\left(e^{-f}\right) \text{ where, } f = \frac{b}{2}\frac{(1-r/R)}{\lambda}$$

$$dT = 4\pi\rho r F (V+v) v dr$$

Resulting Inflow (for hover):

$$\lambda = \sqrt{\left(\frac{\sigma a}{16F}\right)^2 + \frac{\sigma a}{8F}\theta\frac{r}{R}} - \left(\frac{\sigma a}{16F}\right) = \frac{\sigma a}{16F}\left[\sqrt{1+\frac{32F}{\sigma a}\theta\frac{r}{R}} - 1\right]$$

For Climb:

$$\lambda = \sqrt{\left(\frac{\sigma a}{16F} - \frac{\lambda_c}{2}\right)^2 + \frac{\sigma a}{8F}\theta\frac{r}{R}} - \left(\frac{\sigma a}{16F} - \frac{\lambda_c}{2}\right)$$